

Supporting Information for: Has the South American low-level jet changed? Radiosonde evidence and the limits set by the observing system, 1980–2025

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S1 Jet classes and their physics

Definitions.

Between the seasons 1998 and 2025 (October 1997–March 2025), when 925 hPa is reported in most soundings, we separate deep jets (class A) from jets detected only with the full profile (class B, shallow jets) and from days without a jet (class C), and composite the ERA5 southward moisture flux ($-q_v$) at 850 hPa for each class relative to class C. Because the classes are defined by the detection levels, we also classify full-profile jets by the height of their core above the ground (≤ 60 or > 60 hPa), which does not depend on fixed levels, and test the sensitivity of the classes to the speed threshold (11, 12 and 13 m s^{-1}).

Four diagnostics test the physics. First, at the seven Brazilian stations that launch at 00 and 12 UTC (about 20 and 08 local solar time, respectively), we pair the soundings of the same night and compute the change of wind speed at 925 and 850 hPa and the turning of the 925-hPa wind; inertial oscillation theory predicts nocturnal acceleration and a counterclockwise turning that grows with the Coriolis parameter (Blackadar, 1957; Holton, 1967). Second, at the four Argentine airports, we composite the 12-UTC surface pressure and dew point from two days before to three days after each jet, as anomalies from the calendar-month median. Third, we count cold fronts, defined as a 24-h pressure rise of at least 4 hPa together with a 24-h temperature fall of at least 3 °C, in the 48 h after each jet. Fourth, we define regional jet days from the soundings of Resistencia, Córdoba, Santa Rosa and Foz do Iguaçu and compare the CHIRPS rain of the day of the sounding and the next over the exit region of the jet (33–25°S, 62–53°W) as anomalies from the monthly mean; we also regress that anomaly on the class and on synoptic covariates of the jet day (pressure anomaly, 24-h pressure change, and a front within 48 h). Intervals are 90 % intervals from resampling whole seasons; because the nocturnal changes and turning angles are discretized, we report means, not medians. As a check of the large-scale flow we fit a plane to the 12-UTC SLP of the four airports and derive the geostrophic wind.

Results.

Between the seasons 1998 and 2025, jets detected only with the full profile outnumber deep jets by a factor of 2.4 to 5.4 (Resistencia 480 against 199, Córdoba 483 against 189, Santa Rosa 750 against 139). Their core is weaker (14–16 against 17–21 m s^{-1}) and lower (48–69 against 82–90 hPa above the surface, roughly 400–600 m against 700–800 m), and the 850-hPa northerly wind is weaker (7–10 against 14–15 m s^{-1}). In composite profiles of 2006–2025, the median wind of shallow jets reaches 12.9 m s^{-1} at 60 hPa above the surface, against 16.7 m s^{-1} at 100 hPa for deep jets and 7.2 m s^{-1} in the absence of a jet (Fig. S1a). They are not just noise around the threshold: the ratio of shallow to deep jets is 3.3, 3.4 and 3.5 for thresholds of 11, 12 and 13 m s^{-1} , the median core of shallow jets is 14.9 m s^{-1} and only 17–22 % of them lie within 1 m s^{-1} of the threshold. The southward moisture-flux anomaly at 850 hPa is smaller for shallow jets (+68 against +84 $\text{g kg}^{-1} \text{ m s}^{-1}$ over 20–32°S, 65–55°W for Resistencia; +20 against +34 there for Córdoba; +27 against +46 over 30–40°S, 68–58°W for Santa Rosa; Fig. S2c, d, which shows the mean of the three composites).

Nocturnal character. The Brazilian stations launch at 00 UTC (about 20 local solar time, shortly after the collapse of the mixed layer) and at 12 UTC (about 08 local solar time, at the end of the night). Shallow jets are 2 to 11 times more frequent at 12 UTC than at 00 UTC (Foz do Iguaçu 17.1 % against 1.8 %; Uruguaiiana 9.1 % against 2.6 %), and deep jets 2 to 37 times; classified by core height in 2006–2025, jets with a low core (≤ 60 hPa) are 3.4 times and those with a high core 5.4 times more frequent at 12 than at 00 UTC. In nights ending in a shallow jet the wind accelerates by a mean of 4.4 m s^{-1} at 925 hPa (90 % interval 4.2 to 4.8) and 3.0 m s^{-1} at 850 hPa; in nights ending in a deep jet the largest acceleration is at 850 hPa (7.3 m s^{-1}); in nights without a jet it is 0.6 m s^{-1} . Part of this contrast is by construction, because the classes are defined by the 12-UTC wind; the comparison of frequencies at the two hours, which uses the same criterion at both hours, is not.

Inertial turning. When the residual layer decouples from surface friction at sunset, the ageostrophic wind rotates at the inertial frequency, counterclockwise in the Southern Hemisphere (Blackadar, 1957; Holton, 1967). Between 00 and 12 UTC, in nights ending in a shallow jet, the 925-hPa wind backs on average by 34° (31° to 37°) pooled over the seven stations, and by about 50° at Foz do Iguaçu (25.6°S) and Uruguaiiana (29.8°S), 30° at Campo Grande (20.5°S), 20° at Londrina (23.3°S) and 15° at Cuiabá (15.6°S), and does not turn at Vilhena (12.7°S) and veers by 10° at Porto Velho (8.8°S). The turning grows with latitude (Fig. S1d), but it amounts to only 13–20 % of the inertial rotation $f \cdot 12 \text{ h}$ (median over stations; range among stations 0–35 %) in all three classes, and the turning of nights without a jet is not smaller. The turning shows that the nocturnal boundary layer behaves inertially in all nights; it does not by itself distinguish jet nights, and we do not use it as evidence that shallow jets are real.

Synoptic setting. Surface observations at 12 UTC at the four Argentine airports place both jet classes in the classical prefrontal sequence of the jet (Salio et al., 2002; Seluchi et al., 2003; Saulo et al., 2007; Fig. S1b, c). The change of surface pressure in the 24 h before a jet day (deep jets, shallow jets,

no jet) is -2.5 , -2.7 and $+0.4$ hPa at Resistencia, -4.4 , -3.7 and $+0.9$ at Córdoba, -4.0 , -3.0 and $+0.8$ at Santa Rosa and -3.8 , -3.2 and $+0.4$ at Ezeiza. On the day of the jet the pressure anomaly averaged over the four stations is -3.6 hPa for deep jets and -1.6 hPa for shallow jets, temperature is 2.9 and 2.0 °C above normal and dew point 1.4 and 0.3 °C above normal; shallow-jet composites start from a positive pressure anomaly ($+1.4$ and $+1.5$ hPa two and one days before). A cold front follows within 48 h after 53 % of deep jets, 32 % of shallow jets and 10 % of other days at Resistencia, after 42, 33 and 18 % at Córdoba and after 34, 23 and 16 % at Ezeiza; at Santa Rosa the three classes do not differ (24, 20 and 21 %). An alternative definition of the front based on a southerly 850-hPa wind in the soundings of the following 48 h does not separate the classes (36, 32 and 48 % at Resistencia; 62, 50 and 45 % at Córdoba), because southerly wind aloft also follows many days without a jet; we report the surface definition because it does not depend on the soundings.

Rain. Rain in the exit region on the day of a regional jet and the next (two daily CHIRPS totals, which include the 12 h before the 12-UTC sounding) is 14.4 mm after deep jets (90 % interval 12.7 to 16.1), 9.8 mm (9.0 to 10.7) after shallow jets and 7.7 mm (7.1 to 8.4) after days without a jet; 20 mm or more fall after 33, 19 and 12 % of those days. As anomalies from the monthly mean they are $+5.0$ (3.4 to 6.6), $+0.5$ (-0.3 to $+1.3$) and -1.6 mm (-2.3 to -0.9); the rain after deep jets exceeds that after days without a jet by 6.6 mm (5.1 to 8.0) and after shallow jets by 2.1 mm (1.2 to 3.0), and the difference is the same up to 2015 (6.8 and 2.4 mm; Fig. S2a, b). Jet days are associated with fronts, but the association remains when this is accounted for: with a front within 48 h the anomalies are $+8.2$, $+4.6$ and $+1.2$ mm, and without one $+2.4$, -1.6 and -2.4 mm (deep, shallow and no jet); and in a regression on the class and on the pressure anomaly, the 24-h pressure change and the front, the coefficients are $+4.7$ mm (3.5 to 5.9) for deep jets and $+2.3$ mm (1.6 to 3.0) for shallow jets. Classified instead by core height, jets with a low core (≤ 60 hPa) have no rain anomaly ($+0.0$ mm relative to days without a jet, -0.8 to $+1.0$), whereas jets with a core above 60 hPa have $+4.7$ mm (3.7 to 5.7). Deep jets are 13 % of the regional days evaluated and shallow jets 35 %; they account for 21 and 36 % of the sum of the two-day rain totals. We describe these as associations: the synoptic covariates reduce but do not remove the possibility that the larger forcing of deep jets explains their rain.

Shallow jets are thus nocturnal low-level wind maxima, not just threshold noise, that occur in the same prefrontal setting as deep jets with weaker forcing and a smaller rain anomaly, and those with the lowest cores are not associated with rain at all. They resemble the shallow jets observed during the RELAMPAGO campaign (Sasaki et al., 2022).

ERA5 and the nocturnal acceleration. In nights at the Brazilian stations that end in a jet, ERA5 reproduces the inertial turning of the 925-hPa wind (-35° against -34° for shallow jets) but underestimates the overnight acceleration: 3.0 against 4.4 m s^{-1} at 925 hPa for shallow jets, 2.9 against 4.1 at 925 hPa and 5.3 against 7.3 at 850 hPa for deep jets. Conditioning on observed jets can

create such a deficit by regression to the mean, so we checked it two ways. When the nights are selected by the ERA5 jet instead, the sounding still accelerates more than ERA5 (+5.2 against +4.4 m s⁻¹ at 925 hPa), which regression to the mean cannot produce. Without any conditioning, the upper quantiles of the acceleration are lower in ERA5 by 20–25 % (90th percentile 4.8 against 6.2 m s⁻¹; 99th percentile 8.9 against 11.3 m s⁻¹), and those of the 925-hPa wind at 12 UTC by 8–13 % (99th percentile 15.8 against 17.5 m s⁻¹). The deficit has a part due to noise of the soundings and to the difference between a grid cell and a point profile that we cannot separate from a model deficit, so we do not attribute it to the mixing of stable boundary layers in the ECMWF model (Sandu et al., 2013), although it is a candidate. Detection by ERA5 does not depend on the height of the jet core (42 % for cores at or below 60 hPa above the ground, 44 % above), but it is much higher for deep than for shallow jets (78 against 26 % at Córdoba and 47 against 17 % at Campo Grande in 1998–2025), which is consistent with shallow jets lying closer to the detection threshold.

S2 Can published jet trends be checked against soundings?

The largest increases in jet frequency reported in the literature come from regions where long radiosonde records are scarcest. At the two sites used by Montini et al. (2019), Santa Cruz de la Sierra and Mariscal Estigarribia, the IGRA stations within 1.5° have 1454 soundings ending in 2004 and 708 soundings in 2001–2005. In the northern branch, where Jones (2019) found the strongest increases, the network thinned precisely during the period of the reported change: Maracay (Venezuela) reported 1833 soundings at 12 UTC in the 1980s and 166 in the 2010s (all months of the year), and stopped in 2014; San Antonio del Táchira went from 1464 to 386 and stopped in 2013; and Las Gaviotas (Colombia) stopped in 2003. These sparse records show higher frequencies of strong low-level wind maxima after 1990 (fixed-level criterion without direction restriction, because the jet is easterly or northeasterly there: 0.4 to 1.3 % at Maracay, 2.2 to 4.2 % at San Antonio del Táchira and 11.1 to 31.1 % at Las Gaviotas), but too few seasons have at least 60 soundings for a trend (5 to 13 seasons). The long records at Curaçao and Trinidad, under the Caribbean trade-wind regime, show no significant change (+0.3 and +0.5 pp decade⁻¹, $p = 0.55$ and 0.10), and Boa Vista, with a gap in the 1990s, a non-significant decrease (−1.7, $p = 0.15$, 22 seasons). Reanalysis trends of the northern branch therefore cannot be verified with soundings,. We did not reproduce the gridded indices of those studies. Over the La Plata basin, where long soundings exist, the full-profile frequency increased by 7–14 % decade⁻¹ and that of deep jets did not change detectably (Sections 4.2 and 4.3).

S3 Tables

Table S1. Stations used. "12-UTC record" and "12-UTC soundings" refer to soundings at 12 UTC in October–March 1980–2025; the IGRA records of several stations begin earlier at other hours. Used in: T, trend by station (Fig. 1); P, pooled analysis of ten stations; E, comparison with ERA5; B, paired

00 and 12 UTC soundings (Section 4.6); N, northern branch (Supplement S2). The median number of levels is for the lowest 250 hPa above the surface.

IGRA ID	Station	Lat. (°)	Lon. (°)	Elev. (m)	12- UTC record	12-UTC soundings	Levels 1980– 95	Levels 2005– 25	Used in
UCM00078988	Hato Airport; Curacao	12.2	-68.97	8	1980– 2025	6333	6	8	N
COM00080035	Riohacha/Almirante Padilla	11.53	-72.93	4	1992– 2016	1444	4	4	N
TDM00078970	Piarco Int. Airport; Trinidad	10.59	-61.34	12	1980– 2025	7150	8	8	N
VEM00080413	Maracay - B.A. Sucre	10.25	-67.65	436	1980– 2014	1597	6	3	N
VEM00080447	San Antonio Del Tachira	7.85	-72.45	377	1980– 2013	1179	4	3	N
FGM00081405	Cayenne Matoury	4.82	-52.37	8	1980– 2025	5840	4	6	T
COM00080222	Bogota/Eldorado	4.7	-74.15	2547	1980– 2025	6466	7	7	–
COM00080241	Las Gaviotas	4.55	-70.92	167	1980– 2003	1161	5	–	N
BRM00082022	Boa Vista (Aero)	2.85	-60.69	83	1980– 2025	3691	4	8	N
BRM00082193	Belem (Aero)	-1.38	-48.48	16	1982– 2025	6435	7	8	T
BRM00082244	Santarem (Aero)	-2.43	-54.72	72	2006– 2025	3195	–	8	–
BRM00082281	Sao Luis (Aero)	-2.6	-44.23	53	1980– 2021	3501	4	8	–
BRM00082332	Manaus (Aero)	-3.15	-59.98	84	1980– 2025	6352	4	7	T
BRM00082599	Natal (Aero)	-5.92	-35.25	52	1980– 2025	5176	6	8	T
BRM00082824	Porto Velho (Aero)	-8.77	-63.92	102	1980– 2025	4937	4	7	T P E B

IGRA ID	Station	Lat. (°)	Lon. (°)	Elev. (m)	12- UTC record	12-UTC soundings	Levels 1980– 95	Levels 2005– 25	Used in
BRM00082983	Petrolina (82983-0)	-9.38	-40.48	370	1980– 2020	3570	5	7	–
PEM00084628	Lima/Callao	-12.02	-77.12	12	1980– 2020	1906	10	7	–
BRM00083208	Vilhena (Aero)	-12.7	-60.1	612	1980– 2025	4880	6	8	T P E B
BRM00083362	Cuiaba (Aero)	-15.65	-56.1	187	1996– 2021	3565	–	7	T P E B
BRM00083378	Brasilia (Aero)	-15.87	-47.92	1061	1980– 2025	7363	5	7	T
BRM00083566	Confins (Aero)	-19.62	-43.97	827	1994– 2025	4558	5	7	T
BRM00083649	Vitoria (Aero)	-20.27	-40.28	4	2008– 2025	2976	–	8	–
BRM00083612	Campo Grande (Aero)	-20.47	-54.67	567	1980– 2025	5920	3	8	T P E B
BRM00083746	Galeao (Aero)	-22.82	-43.25	6	1980– 2024	7421	6	7	T
BRM00083768	Londrina (Aero)	-23.33	-51.13	569	2007– 2025	2975	–	8	E B
CIM00085442	Antofagasta	-23.45	-70.45	112	1980– 2025	6620	8	7	T
BRM00083840	Curitiba (Aero)	-25.52	-49.17	908	1980– 2025	6831	5	7	T
BRM00083827	Foz Do Iguacu (Aero)	-25.6	-54.48	239	1992– 2025	4530	11	8	T P E B
ARM00087155	Resistencia Aero	-27.44	-59.05	53	1980– 2025	7313	7	8	T P E
BRM00083899	Florianopolis (Aero)	-27.67	-48.55	5	2003– 2025	3526	–	8	T
BRM00083928	Uruguaiana (Aero)	-29.78	-57.03	74	2004– 2025	2876	–	8	P E B

IGRA ID	Station	Lat. (°)	Lon. (°)	Elev. (m)	12- UTC record	12-UTC soundings	Levels 1980– 95	Levels 2005– 25	Used in
ARMooo87344	Cordoba Aero	-31.3	-64.21	493	1980– 2025	6839	5	8	T P E
ARMooo87418	Mendoza Aero	-32.84	-68.8	705	1980– 2025	5197	5	7	T
ARMooo87576	Ezeiza Aero	-34.82	-58.54	20	1980– 2022	7354	6	8	T P E
ARMooo87623	Santa Rosa Aero	-36.59	-64.28	190	1980– 2025	6966	6	7	T P E
ARMooo87715	Neuquen Aero	-38.95	-68.14	270	1980– 2025	4524	5	7	T
CIMooo85799	Puerto Montt	-41.44	-73.1	87	1980– 2025	7853	5	9	T

Table S2. Breaks of the sounding-minus-ERA5 wind speed detected at each station (step: mean change in the six years after minus the six years before; control: mean change at the stations of the other country in the same windows; a break is regional, and not adjusted, if the control moves in the same direction by at least half as much).

Level (hPa)	Station	Date	Step (m s ⁻¹)	Control (m s ⁻¹)	Control stations	Decision
925	Campo Grande	2003-11	0.54	0.36	4	regional, not adjusted
925	Ezeiza	1997-10	0.8	0.01	2	adjusted
925	Foz	2010-02	0.83	-0.01	4	adjusted
925	Resistencia	1997-11	1.12	-0.05	2	adjusted
925	Santa Rosa	2000-02	0.78	0.01	3	adjusted
850	Campo Grande	2000-03	-0.42	0.28	4	adjusted
850	Ezeiza	1992-10	-0.61	0.16	2	adjusted
850	Ezeiza	1997-10	0.09	0.21	4	regional, not adjusted
850	Resistencia	1990-01	-1.19	0.35	2	adjusted

Level (hPa)	Station	Date	Step (m s ⁻¹)	Control (m s ⁻¹)	Control stations	Decision
850	Resistencia	1998-01	0.74	0.17	4	adjusted
850	Resistencia	2018-11	-0.29	-0.03	6	adjusted
850	Santa Rosa	1991-10	-1.0	0.13	2	adjusted
850	Santa Rosa	1997-12	0.44	0.17	4	adjusted
850	Santa Rosa	2017-12	-0.23	-0.03	6	adjusted
700	Campo Grande	1996-10	-0.4	0.09	4	adjusted
700	Campo Grande	2005- 12	-0.45	-0.03	4	adjusted
700	Cuiabá	2009- 01	-0.29	-0.27	4	regional, not adjusted
700	Uruguaiana	2010- 02	-0.23	-0.23	4	regional, not adjusted

S4 Figures

Figure S1. Characteristics of deep (850/700 hPa), shallow (full profile only) and no-jet days. (a) Composite wind profile (median and interquartile range) against height above the ground, 2006–2025. (b) Surface pressure and (c) dew-point anomalies at 12 UTC from two days before to three days after the sounding, mean of four Argentine airports, with 90 % intervals from season resampling. (d) Mean turning of the 925-hPa wind between 00 and 12 UTC against latitude at seven Brazilian stations with 90 % intervals, and the inertial rotation $f \cdot 12$ h.

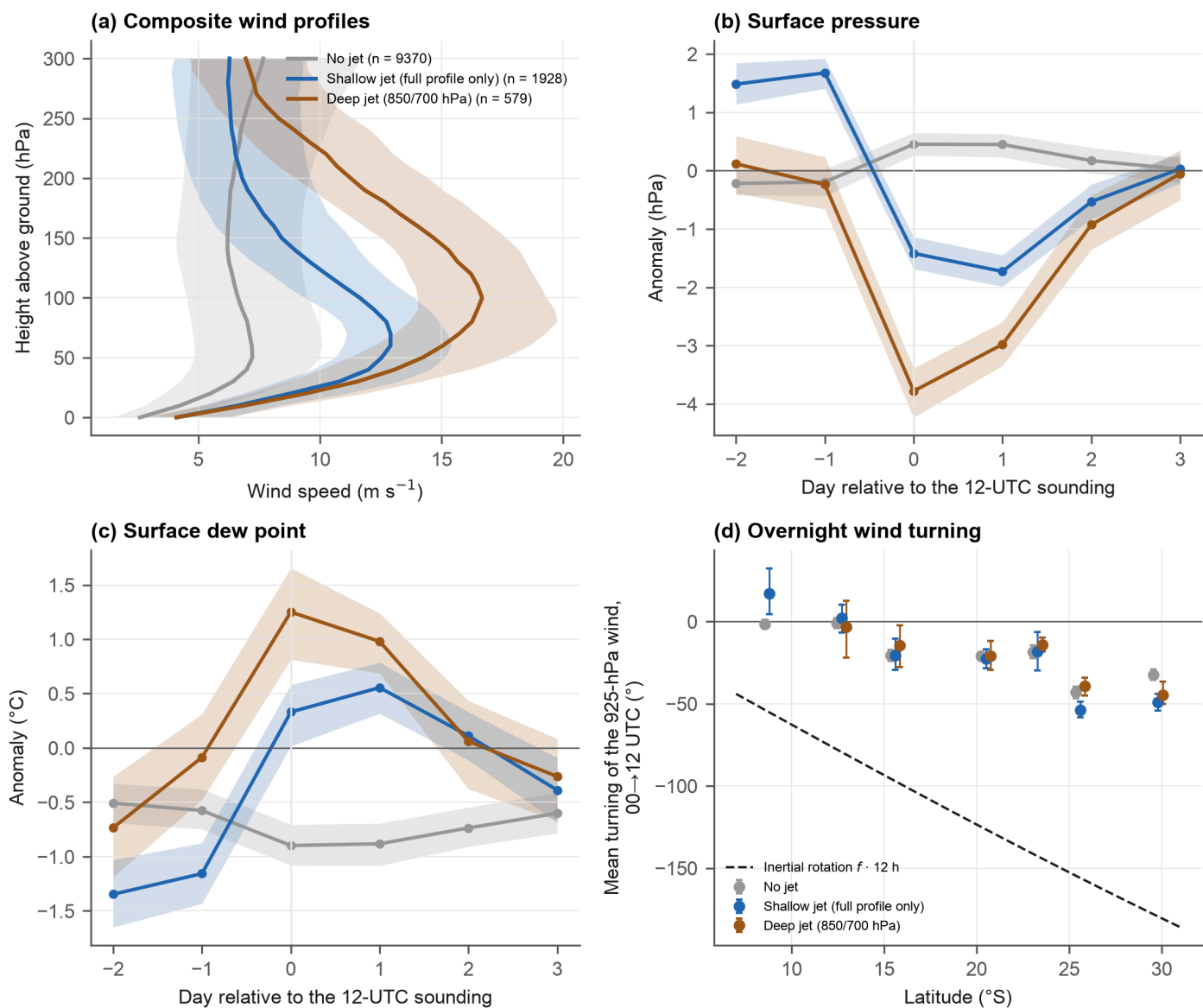
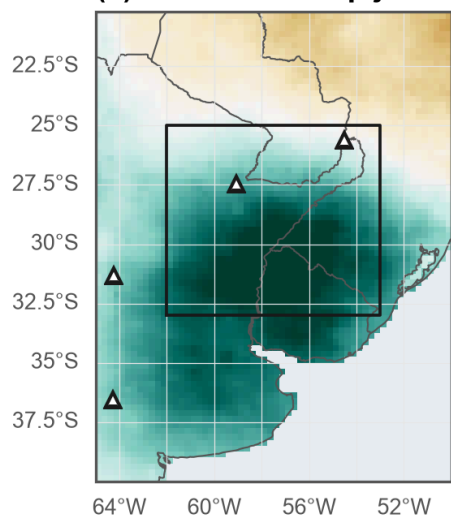
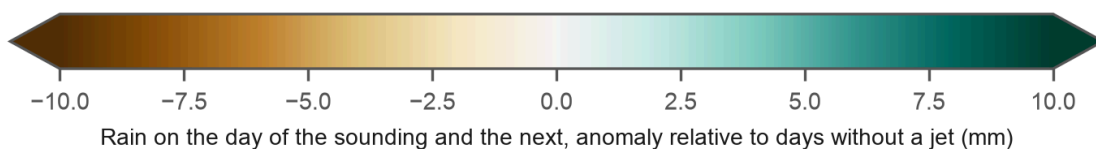
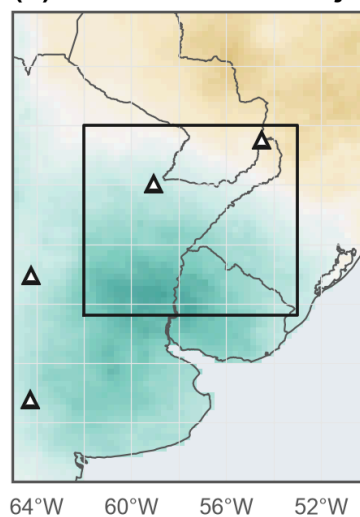


Figure S2. (a, b) CHIRPS rain on the day of deep and shallow regional jets and the next minus the mean after days without a jet, October 1997–March 2025; the box is the exit region used in the text. (c, d) ERA5 southward moisture flux at 850 hPa on deep and shallow jet days, mean of the composites of Resistencia, Córdoba and Santa Rosa, minus days without a jet (solid box: exit region used for Resistencia and Córdoba; dashed box: for Santa Rosa). Triangles: radiosonde stations. No pointwise significance test is shown.

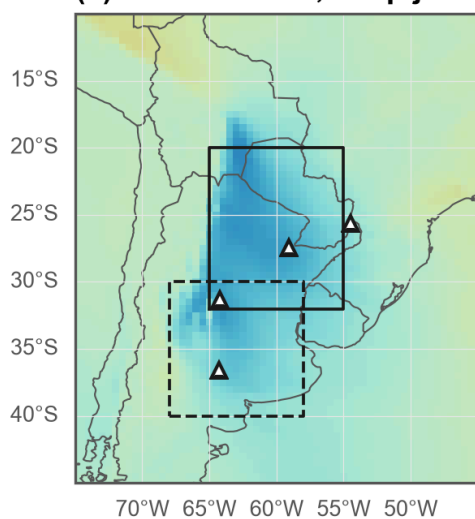
(a) Rain after deep jets – no jet



(b) Rain after shallow jets – no jet



(c) Moisture flux, deep jets



(d) Moisture flux, shallow jets

